Ayush Kumar

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Portfolio | LeetCode: 200+ Solved | GeeksforGeeks | Bhubaneswar, Odisha, India

Summary

Final-Year B.Tech Information Technology student at IIIT Bhubaneswar with strong foundations in Data Structures & Algorithms, Full-Stack Software Engineering (Java, MERN, Python), and Applied Generative AI/LLM Systems. Experienced in architecting scalable RESTful APIs, autonomous agent workflows (LangGraph), vector search pipelines (pgvector), and containerized microservices (Docker, CI/CD). Actively seeking Full-Time Software Development Engineer (SDE) and Graduate Engineer roles.

Education

International Institute of Information Technology (IIIT) — **Bhubaneswar, Odisha**
Bachelor of Technology in Information Technology (CGPA: 7.48) — *Expected May 2026*

Delhi Public School, Nalco Nagar — **Angul, Odisha**
Class XII (Senior Secondary): 88.6% | *Class X (Secondary)*: 98.2% — 2020 – 2022

Technical Skills

Languages:	Java (Core Java, OOP, Collections), JavaScript (ES6+), Python, TypeScript, SQL, C/C++, HTML5, CSS3
Web Technologies:	React.js, Node.js, Express.js, FastAPI, Flask, RESTful APIs, Microservices, Vite, Tailwind CSS, Bootstrap 5
AI, Machine Learning & GenAI:	LangGraph, Autonomous Agents, RAG Pipelines, pgvector, Scikit-learn, XGBoost, Pandas, NumPy, Google Gemini API
Databases & DevOps:	PostgreSQL 16, MongoDB Atlas, Docker, Docker Compose, Git, GitHub Actions (CI/CD), Linux, Render, Cloudinary
Core CS Fundamentals:	Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Operating Systems, System Design

Projects

AgentIQ | *Python, FastAPI, LangGraph, PostgreSQL, pgvector, React, TypeScript, Docker Compose* — GitHub | Live Demo
- Architected an enterprise agentic RAG platform using FastAPI and LangGraph, orchestrating bounded multi-step reasoning (≤ 4 steps) with deterministic fallbacks, execution timeouts, and sub-second retrieval latency.
- Implemented hybrid semantic search via PostgreSQL 16 and pgvector (HNSW indexing, 0.45 cosine cutoff) with source citations, protected by a 5-layer SQL guardrail with sqlglot AST parsing and RBAC.
- Constructed an enterprise-grade 3-pane UI in React 18 and TypeScript, containerizing full-stack services with multi-stage Docker and establishing automated GitHub Actions CI/CD testing.

Feasto | *React.js, Node.js, Express.js, MongoDB Atlas, Razorpay, Google Gemini API, Docker* — GitHub | Live Demo
- Engineered a production-grade food delivery ecosystem following MERN architecture with decoupled customer and admin management portals, featuring real-time driver delivery tracking with Leaflet GPS and persistent cart state.
- Integrated Razorpay Payment Gateway with cryptographic HMAC-SHA256 webhook signature verification, ensuring zero transaction discrepancies across shopping carts, order lifecycles, and database states.
- Embedded an AI food concierge using Google Gemini API for personalized dish recommendations; implemented JWT role-based access control (RBAC), Cloudinary CDN image pipelines, and multi-container Docker deployment.

StayNest | *Node.js, Express.js, MongoDB Atlas, EJS, Bootstrap 5, Cloudinary, Razorpay, PDFKit* — GitHub | Live Demo
- Developed a full-stack accommodation marketplace adopting MVC architecture with full CRUD listings, geospatial filters, dynamic reviews, and responsive UI built with Bootstrap 5.
- Engineered a collision-proof booking engine featuring interactive blackout calendar scheduling, Razorpay payment verification, and automated server-side PDF invoice generation using PDFKit.
- Hardened backend application security with Passport.js session authentication, Joi input validation schemas, Helmet HTTP security headers, and MongoDB Atlas cloud storage.

Smart House Price Predictor | *Python, Flask, XGBoost, Scikit-learn, Pandas, NumPy, Docker* — GitHub | Live Demo
- Developed a California real estate valuation engine achieving R^2 = 83.93% and 17.52% MAPE ($30,245 MAE) using an optimized XGBoost gradient-boosted ensemble trained on 20,000+ property records.
- Integrated distribution-free Split Conformal Prediction delivering mathematically guaranteed 90% confidence intervals (±$68,169), eliminating financial tail risk compared to traditional point estimates.
- Implemented Explainable AI (XAI) feature attribution telemetry and defensive input validation, deploying via a low-latency (<20ms) Flask REST API containerized with Docker and automated GitHub Actions CI.

Achievements & Coding Profiles

- Solved **200+ Data Structures & Algorithms** problems in Java across **LeetCode** and **GeeksforGeeks**, demonstrating algorithmic optimization and problem-solving skills.
- Architected, containerized, and deployed **4 production systems** (AgentIQ, Feasto, StayNest, SmartHouse) with Docker, CI/CD pipelines, and cloud hosting on Render.

- Demonstrated sustained academic excellence with **98.2% in Class X** and **88.6% in Class XII**; Final-Year B.Tech IT student at IIIT Bhubaneswar.